PRIME HOSPITALITY BUSINESS GROUP · SIYONAT BRANCH

Hotel Food & Beverage Cost Control

Complete Crash Course, Architectural Logic & Operational Master Manual

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Module 1: The Core Philosophy of Hotel F&B Cost Control (The "Why")

In retail or supermarket accounting, if you buy 100 shirts for \$10 and sell them for \$20, calculating your cost is simple: every shirt sold removes exactly \$10 of inventory.

Food & Beverage is completely different. You buy a whole beef hindquarter, trim off 15% fat, discard 10% bone, portion the prime cut into *Special Tibs*, marinate it with spiced butter (*Niter Kibbeh*), cook it on a stove, serve it with *Injera*, comp portions to VIP guests, feed your cooks on shift, and ring up sales across 171 buttons on a CNet POS terminal.

THE 4 DEADLY LEAKS WHERE HOTEL MONEY VANISHES

1. **Receiving & Price Creep:** Vendors deliver 47 kg instead of 50 kg, or quietly invoice at 1,500 ETB instead of the 1,400 ETB contracted price.
2. **Kitchen Prep & Over-Portioning:** A recipe calls for 200g of beef, but the kitchen serves 280g because they eyeball rather than scale, inflating food cost by 40%.
3. **Unrecorded Comps & Waste:** Spoiled meat, burnt orders, or friend comps that get thrown away without an official waste log or management sign-off.
4. **Bar Pilferage & Free Pouring:** A bartender gives out double shots or pockets cash on cash sales because liquor inventory is not audited against registered POS sales.

1.1 The Classical Materials Control Cost Equation

Materials Control (HMC) solves this through the universal hospitality cost formula, calculated at the close of every shift or month:

```
Gross Food/Beverage Consumption =
    Opening Physical Stock
  + Direct Supplier Purchases
  + Internal Transfers In
  - Internal Transfers Out
  ± Net Sub-Recipe Batch Production
  - Closing Physical Stock

Net Cost of Sales (True Cost) =
    Gross Consumption
  - Staff Duty Meals (Allocated to HR/Labor Expense)
  - Authorized Promotional / VIP Comps (Allocated to Marketing Expense)

Actual Cost % = (Net Cost of Sales ÷ Net POS Revenue) × 100
```

> **THE GOLDEN LAW: FOOD AND BEVERAGE ARE NEVER MERGED**
>
> Food Cost % and Beverage Cost % must **never** be blended together into a single "F&B %". In a well-run hotel, Beverage Cost target is typically **18% – 22%** (high margin, low prep labor), while Food Cost target is typically **30% – 34%** (perishable, high culinary preparation). Blending them blinds management to kitchen theft or bar waste.

1.2 Moving Weighted Average Cost (MAC / GLD) vs Standard Cost

In Materials Control, inventory is valued using **Gleitender Durchschnitt (Moving Weighted Average)**:

- You have 10 kg of beef on hand bought at **1,200 ETB/kg** (Value = 12,000 ETB).
- You receive 10 kg more at **1,400 ETB/kg** (Value = 14,000 ETB).
- Your new Moving Weighted Average = (12,000 + 14,000) ÷ 20 kg = **1,300 ETB/kg**.
- Every dish sold or transferred out from that moment is depleted at 1,300 ETB. The ledger is *forward-only*: yesterday's closed dishes are never retroactively altered.

Module 2: Complete Master Data Setup from Scratch (Clean Slate)

You requested a clean slate (zero sample data) so you can enter your live hotel data yourself. Here is how every tab in the **Setup** menu works:

2.1 Stores & Cost Centers (`/setup/stores`)

A "Store" is any physical room where inventory sits. The system organizes Prime Hospitality into a clear supply hierarchy:

STORE CODE	STORE NAME	ROLE / KIND	FULFILLING SOURCE (SUPPLY ROUTE)
`MAIN`	Central Store (Warehouse)	Main Warehouse	Direct from external suppliers (Addis Abattoir, BGI, etc.)
`KITCHEN`	Main Kitchen	Kitchen Cost Center	Supplied by Central Store (`MAIN`) via Par Requisitions
`BAR`	Main Bar	Bar Cost Center	Supplied by Central Store (`MAIN`) via Par Requisitions
`PREP`	Pastry & Prep Kitchen	Prep Station	Supplied by `MAIN` , produces sub-recipes (Injera, sauces)

2.2 Categories (`/setup/categories`)

Categories group articles by accounting classification. Each category must have a strict `Kind` :

- **Food Categories:** Meat, Dairy, Fresh Produce, Grains & Flour, Spices & Seasonings.
- **Beverage Categories:** Coffee & Tea, Soft Drinks & Water, Beer & Draught, Wines & Spirits.
- **Other:** Disposables, Takeaway packaging, guest amenities.

2.3 Raw Material Items & Dual Units (`/setup/items`)

When entering an ingredient, Materials Control distinguishes between how you **count** it versus how you **buy** it:

FIELD	PURPOSE	REAL EXAMPLE
Inventory UOM	How the kitchen or bar uses and counts the item.	`Kg` for Teff; `Bottle` for Beer; `L` for Milk.
Purchase UOM	How the vendor packages and bills the item.	`Bag` (50 kg); `Crate` (24 bottles); `Box`.
Pack Factor (`pack_factor`)	How many inventory units are inside 1 purchase unit.	**50** (1 bag = 50 kg Teff); **24** (1 crate = 24 bottles of Ambo).
Standard Loss % (Shrinkage)	Expected peeling, trimming, or evaporative yield loss.	**15%** for Onions (peeling/roots); **0%** for sealed bottled soda.
ABC Classification	Inventory capital prioritization (Pareto Rule).	**Class A:** Beef Fillet, Whiskey (Count daily). **Class C:** Salt, Toothpicks (Count monthly).

2.4 Suppliers & Contract Price Books (`/setup/suppliers`)

Register each vendor (*Addis Abattoir Enterprise*, *BGI Ethiopia*, *Merkato Wholesale Grain Traders*). Click on a supplier to open their **Contract Price Book**:

- **Contract Price:** Contracted price in ETB (e.g., 6,500 ETB per 50kg Bag of Teff).
- **Tax Rate:** 15% VAT for commercial beverages/processed goods, or 0% for zero-rated basic grains.
- **Min Order Qty (MOQ):** Minimum units vendor will deliver (e.g., 5 bags).
- **Lead Time:** Delivery turnaround in days (e.g., 2 days).

2.5 Recipes & Sub-Recipes (`/setup/recipes`)

The recipe tree connects raw ingredients to POS sales buttons:

- **1. Sub-Recipes (Prep Kitchen Batches):** Prepared items made in advance:
  Example: **Injera Batch (100 Pcs)** consumes 30 kg Teff flour + yeast + water.
  Example: **Kitfo Base (5 Kg)** consumes 4.5 kg Minced Beef + 0.5 kg Niter Kibbeh + Mitmita.
- **2. Menu Dish Recipes:** The finished dish rung up on the till:
  Example: **Special Tibs** consumes:
  - 0.250 kg Beef Fillet
  - 0.080 kg White Onion
  - 0.030 kg Spiced Butter (*Niter Kibbeh*)
  - 2 pcs Injera (nested sub-recipe)

2.6 POS Article Mapping (`/setup/pos-map`)

When your CNet cash registers print a bill, each button has an Article Code (e.g., `101` for "SPECIAL TIBS"). In this screen, link Article Code `101` to the "Special Tibs" recipe.
Whenever an Excel export has a new button, it automatically appears here under **Unmapped Only** so you can link it without manual entry.

2.7 Par Stock Levels (`/setup/stores/[id]`)

For every outlet, establish two safety numbers for each item:

```
Min Par (Safety Buffer): The absolute minimum stock before the kitchen runs out.
Order-Up-To Par (Max Ceiling): The ideal stock level to hold for a busy service.

Example: Main Kitchen holds Min Par = 10 kg and Max Par = 30 kg of Beef Fillet.
If current on-hand stock is 12 kg, 1-click requisition automatically orders 18 kg (30 -
12) from the Central Store!
```

Module 3: Daily Operational Workflows (A Day in the Life)

Here is how the team runs the hotel using the software throughout a 24-hour cycle:

Workflow 1: Morning Deliveries & 3-Way Matching (Storekeeper)

1. Vendor arrives at the loading dock with physical goods and a delivery note.
2. Storekeeper signs in, opens **Purchases** (`/purchases`), and clicks **+ New Delivery (GRN)**.
3. Selects vendor (e.g. *Addis Abattoir*), enters delivery note number, and inputs physically weighed quantities and invoice prices.
4. **3-Way Matching Gate:** If invoice price exceeds contracted price by more than 2%, the system flags a price variance alert requiring Cost Controller authorization.
5. Clicking **Book Delivery** adds stock to the Central Store and updates moving average costs.

Workflow 2: Internal Requisitions (Chef $\rightarrow$ Storekeeper)

1. Before lunch prep, the Chef opens **Requisitions** (`/requests`).
2. Selects **Main Kitchen** and clicks ⚡ **Auto-Fill from Par Stock**.
3. The system automatically populates replenishment quantities for every ingredient below par. Chef clicks **Submit Requisition**.
4. The Storekeeper opens `/requests` , picks the goods in the warehouse, and clicks **Fulfill & Issue Stock**.
5. Value-neutral movement: Central Store decreases, Main Kitchen increases at exact current moving average cost.

Workflow 3: Butcher Yield Tests & Prep Cooking (Chef)

1. The hotel purchases a whole 100 kg beef carcass at 800 ETB/kg (Gross Cost = 80,000 ETB).
2. Chef breaks it down in the butchery and opens **Production** (`/production`):
    - **Prime Cuts extracted:** 12 kg Tenderloin, 18 kg Ribeye, 20 kg Sirloin, 30 kg Minced Meat.
    - **Byproducts salvaged:** 15 kg Soup Bones (worth 150 ETB/kg) + 5 kg Tallow Fat (worth 200 ETB/kg).
3. **Byproduct Credit Salvage Method:** The system subtracts the 3,250 ETB byproduct value from the 80,000 ETB carcass cost, deriving a net meat cost of 76,750 ETB divided across prime cuts.
4. Clicking **Save & Execute** depletes the carcass and books individual prime cut portion items into the kitchen ledger!

Workflow 4: Waste & Spoilage Logging (Kitchen & Bar)

1. If a cook burns 3 orders of Tibs or a bartender drops a bottle of gin, they open **Waste** (`/waste`).
2. Record the store, item or recipe, quantity, and reason (*burnt, spoiled, expired, broken*).
3. Cost Controller reviews and clicks **Approve Write-off**, deducting inventory and posting to the Waste Loss expense ledger.

Workflow 5: Night Audit — CNet Sales Depletion (Cost Controller)

1. At the close of business, the night auditor exports `cnet_sales_export.xlsx` from the CNet POS server.
2. Cost Controller signs in and opens **Sales & Depletions** (`/sales/eod`).
3. Drags and drops the Excel file. The system runs the **Reconciliation Control Gate**:
   Checks if row sums match the file footer (Gross Revenue, Net Sales, 15% VAT, 10% Service Charge).

4. Controller clicks: **"Explode Recipes & Post Stock Depletions"**.
5. The system calculates all 1,200 dishes sold, cascades through all sub-recipes, and depletes exact grams of beef, onions, butter, and beer from kitchen and bar stores.

Workflow 6: Physical Inventory Counts & Recount Alerts (Auditors)

1. Weekly or monthly, auditors open **Physical Counts** (`/counts`) and click **+ New Count Sheet**.
2. Turn on **Blind Count Sheet Mode** and click **Print Sheet**.
   Why Blind? Theoretical quantities are hidden so counting staff cannot guess or fake counts.
3. Staff physically count shelves and enter quantities into the screen.
4. 🚨 **Recount Threshold Alert:** If any item variance exceeds **500 ETB or 5%**, a high-visibility warning flags that item, requiring a supervisor recount and root-cause classification (*portion_error, theft, unrecorded_waste*).

Module 4: Deciphering the Executive Dashboards & Reports

Everything ties together on the **Executive Cost Control Dashboard** (`/reports/cost-control`):

4.1 The F&B Reconciliation Card

Side-by-side comparison of 🍳 **Food Cost** vs. 🍸 **Beverage Cost**:

- **Theoretical Target Cost %:** What cost *should* have been based on recipes sold.
- **Actual Cost %:** What cost *actually* was based on starting stock + purchases - ending stock.
- **Variance % & Potential Savings (ETB):** The difference represents avoidable operational leakage. If Actual Food Cost is 34% but Theoretical was 29% on 2,000,000 ETB sales, you have **100,000 ETB in potential savings** lost to theft, waste, or over-portioning!

4.2 Cost Center Performance Matrix

Outlines each outlet's operational velocity:

- **Deliveries & Transfers In/Out:** Exact value of stock flowing through each kitchen and bar.
- **Turnover Ratio (x):** Measures stock velocity (Usage ÷ Average Inventory). High turnover means fresh, lean stock; low turnover means cash trapped in dead inventory.

4.3 Pareto ABC Inventory Capital Analysis

Categorizes property inventory according to capital investment:

- **Class A (80% of property capital):** Meat, Primal cuts, Imported spirits. *Policy: Counted daily or weekly.*
- **Class B (15% of property capital):** Dairy, Grains, Beer crates. *Policy: Counted bi-weekly.*
- **Class C (5% of property capital):** Spices, Salt, Napkins, Baking powder. *Policy: Counted monthly, ordered in bulk.*

4.4 Article Stock Card (Artikelkarte) Audit Trail

An immutable forward ledger for any article (e.g. Beef Fillet) in any store:
Shows Date, Transaction Type, Source Document, Qty Change, Unit Cost, and running balance. If 5 kg goes missing, you can trace the exact timestamp and user responsible.

Module 5: Staff Shift Checklist & Onboarding

ROLE	MORNING SHIFT	AFTERNOON SHIFT	CLOSING / NIGHT AUDIT
Central Storekeeper	Receive morning produce and meat deliveries (GRN). Check price variances against contract.	Fulfill and dispatch pending outlet Par requisitions to Kitchen and Bar.	Lock warehouse, review zero-balance items, generate vendor PO replenishment.
Executive Chef	Inspect walk-in cooler, submit 1-click Par Requisition on `/requests`.	Log carcass breakdown yield tests on `/production`. Execute prep batches (Injera, kibbeh).	Record kitchen prep waste on `/waste`. Conduct Class A meat spot-check.
Bar Manager	Submit beverage replenishment requisition from Central Store.	Accept deliveries, verify bottle seals and crate counts. Record any breakage.	Count high-value open liquor bottles. Lock liquor cage.
Cost Controller / GM	Review yesterday's delivery price variances and approve write-offs.	Conduct random blind spot-checks in outlets. Calibrate chef recipe costs.	Upload daily CNet sales export on `/sales/eod`, explode recipes, and review Cost % Dashboard.

CLEAN SLATE RESET REMINDER

Whenever you want to wipe the slate clean and start fresh with zero demo articles, run in your terminal:
`npm run db:bare`
This keeps your Prime Hospitality stores and user accounts intact, ready for your real data entry!